VLSI Testing: 2009 Midterm

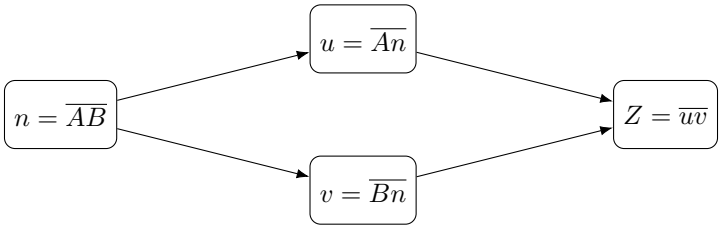
Reconstructed questions and independently derived solutions. Personal names, identifiers, handwritten answers and administrative headers are omitted. Figures are redrawn as logical diagrams or explicit netlists. A source-reading note identifies a meaningful ambiguity instead of silently changing the question. Finite Boolean checks are supplied in the examination collection's Python verification script.

1 SCOAP for two combinational circuits

Question. Compute controllability and observability for (a) the four-NAND XOR realization and (b) the AND–OR mux realization. Input names below replace unlabeled input wires.

Use $CC_0 = CC_1 = 1$ at a primary input and $CO = 0$ at a primary output; each gate adds one. For an AND gate, $CC_0 = 1 + \min CC_0(i)$ and $CC_1 = 1 + \sum CC_1(i)$. For an OR gate, interchange the roles of 0 and 1. NAND/NOR swap the two output controllabilities; an inverter swaps its input values and adds one. To observe one gate input, set the other inputs to their noncontrolling values, add their controllability costs, add CO of the output and add one. At a fanout stem take the minimum branch CO. For XOR/XNOR, minimize the sum of input costs subject to the desired parity. These are structural SCOAP estimates; reconvergent inputs are treated independently.

Signal	Defining expression
n	$\overline{AB}$
u	$\overline{An}$
v	$\overline{Bn}$
Z	$\overline{uv}$



Signal	CC_0	CC_1	CO
A	1	1	6
B	1	1	6
n	3	2	5
u	4	2	3
v	4	2	3
Z	5	5	0

The A, B stems each have $CO = 6$; their branches into the first NAND have $CO = 7$. Branches into the second level have $CO = 6$.

Signal	Defining expression
n	$\bar{S}$
u	AS
v	Bn
Z	$u + v$

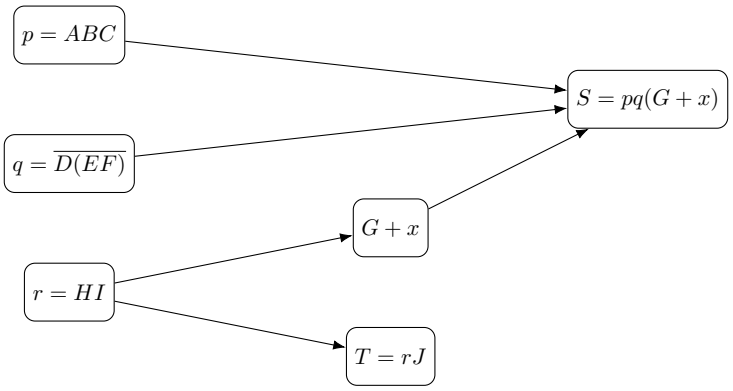
Signal	CC_0	CC_1	CO
A	1	1	5
S	1	1	5
B	1	1	6
n	2	2	5
u	2	3	3
v	2	4	3
Z	5	4	0

For (b), $CO(S \to u) = 5$ and $CO(S \to n) = 6$; take $CO(S) = 5$. The costs in (a) are structural estimates despite the function reducing to XOR.

2 ATPG and pseudo-exhaustive testing

Question. For the following circuit: (a) find a D-algorithm test for $x/0$; (b) find a single-path test for $y/0$; (c) determine a pseudo-exhaustive test count. The source also has a handwritten correction of (b) to $y/1$.

Signal	Defining expression
p	ABC
y	EF
q	$\overline{Dy}$
r	HI
x	r branch to the OR gate
S	$pq(G + x)$
T	rJ



Solution. For $x/0$, set $H = I = 1$ so $x = D$, $G = 0$ to propagate through OR, $A = B = C = 1$ and $D = 0$ to make $p = q = 1$. For example $(A, B, C, D, E, F, G, H, I, J) = 1110000110$ gives $S = 1/0$. For the printed $y/0$, set $E = F = D = 1$, $A = B = C = 1$, $G = 1$. Test 1111111000 makes

the NAND and S change from 0 to 1. For the handwritten $y/1$, test 1111001000 gives $S = 1/0$. Thus both readings are solved rather than silently replacing the printed fault.

The S cone depends on nine inputs $A, \ldots, I$. Exhaust their $2^9 = 512$ combinations. Choose $J = A$; then (H, I, J) also covers all eight combinations of the T cone. This is a sufficient conventional pseudo-exhaustive construction, not a proof of a minimum compressed test count.

3 Equivalence and dominance

Question. For $d = \overline{ab}$ and $e = \overline{d + c}$, classify $a/0$, $b/1$, $d/1$ and $e/0$ by equivalence and dominance.



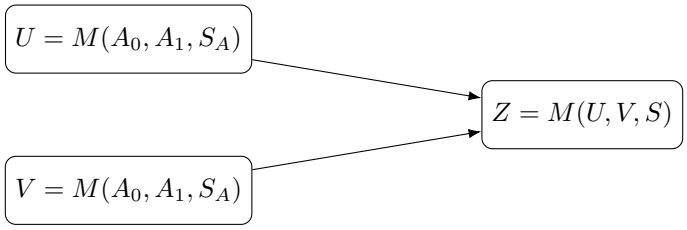
Solution. Since $e = ab\bar{c}$, the detection sets, in (a, b, c) order, are

$$T(a/0) = T(d/1) = T(e/0) = \{110\}, \qquad T(b/1) = \{100\}.$$

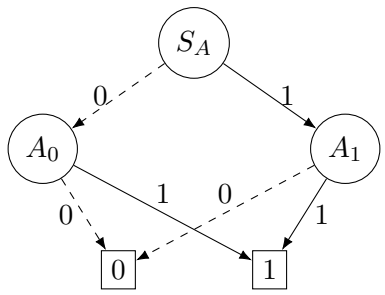
Thus the first three faults are equivalent. The fourth is neither equivalent to nor in a strict dominance relationship with them: the two nonempty detection sets are disjoint. Stating the sets also avoids opposite conventions for dominance.

4 BDD and tests for a mux network

Question. Two identical muxes with data A_0, A_1 and select S_A feed a final mux selected by S. (a) Build a BDD using one 0 leaf and one 1 leaf. (b) Find all tests for the primary-output stuck-at faults. The source crosses out *outputs* and writes *inputs*; solve both.



Solution. The final data inputs are equal: $Z = \bar{S}_A A_0 + S_A A_1$, independent of S. A reduced ordered BDD with order S_A, A_0, A_1 is:

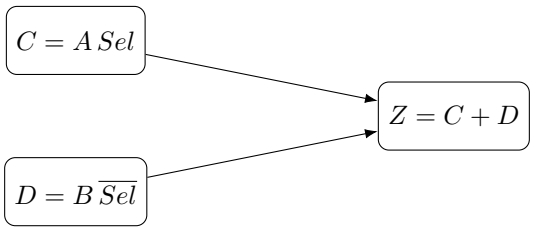


Output-fault tests are $T(Z/0) = \{Z = 1\}$ and $T(Z/1) = \{Z = 0\}$. For input stem faults, with unmentioned variables arbitrary:

Fault	All detecting assignments
$A_0/0$	$S_A = 0, A_0 = 1$
$A_0/1$	$S_A = 0, A_0 = 0$
$A_1/0$	$S_A = 1, A_1 = 1$
$A_1/1$	$S_A = 1, A_1 = 0$
$S_A/0$	$S_A = 1, A_0 \neq A_1$
$S_A/1$	$S_A = 0, A_0 \neq A_1$
$S/0,\ S/1$	no tests (redundant select)

5 A hidden gate-level mux fault

Question. A mux is realized by $C = A\,Sel$, $n = \overline{Sel}$, $D = Bn$, $Z = C + D$. Find a functional-block pin test set covering all pin SA0/SA1 faults but missing $n/1$ at the input of AND gate D.



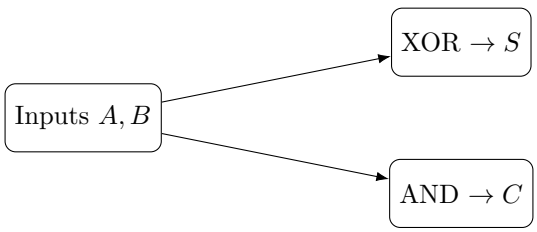
Solution. In (A, B, Sel) order use $\{000, 001, 010, 101\}$, with outputs $0, 0, 1, 1$. $A/0$ is detected by 101, $A/1$ by 001, $B/0$ by 010, $B/1$ by 000, $Sel/0$ by 101 and $Sel/1$ by 010. Both output faults are covered. However $n/1$ is detected only by $Sel = 1, B = 1, A = 0$, namely 011, which is absent. Functional pin coverage does not imply internal structural coverage.

6 XOR deductive simulation and a half-adder

Question. Give a general XOR deductive rule and simulate $S = A \oplus B$, $C = AB$ under inputs 01 and 10.

Solution. A fault in an odd number of XOR-input fault lists changes parity. Hence $L(S) = L(A) \triangle L(B)$, followed by insertion of the locally excited output stuck-at fault. For m inputs take the symmetric difference of all m lists. A fault common to two lists cancels. With signal-stem SA0/SA1 faults:

Input	$L(S)$	$L(C)$
01	$\{A/1, B/0, S/0\}$	$\{A/1, C/1\}$
10	$\{A/0, B/1, S/0\}$	$\{B/1, C/1\}$



7 Short questions on manufacturing test

(a) **Verification and test.** Verification asks whether the design implements the specification. Manufacturing test applies stimuli to fabricated hardware and checks for physical defects represented by a fault model.

(b) **Yield and defect level.** With yield $Y = 0.5$, use the model $DL = 1 - Y^{1-FC}$ for the defect fraction among shipped parts. For $DL \leq 100 \times 10^{-6}$,

$$FC \geq 1 - \frac{\ln(0.9999)}{\ln(0.5)} = 0.99985572\ldots,$$

so at least 99.9856% fault coverage is required *under this model*. Yield alone cannot determine defect level without a coverage/escape model.

(c) **Test flow.** Select a defect/fault model; generate and simulate tests; prepare tester programs and expected responses; apply tests at wafer/package/board stages as appropriate; compare responses, diagnose failures and bin devices; feed results back to design and manufacturing.

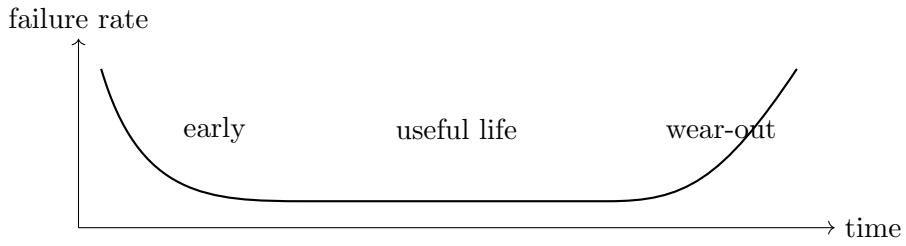
(d) **Detection, location, diagnosis.** Detection distinguishes faulty from fault-free behavior. Location narrows the physical or logical site. Diagnosis infers likely fault mechanisms/sites from the response syndrome.

(e) **Fault modeling.** A finite logical/electrical abstraction of defects makes test generation and coverage measurable; model coverage need not equal defect coverage.

(f) **Bathtub curve.** Early failures decrease as weak parts fail; useful life has a low approximately constant hazard; wear-out failures increase with aging.

(g) **Fault coverage.** $FC = N_{\text{detected}}/N_{\text{modeled}}$, with the denominator and any exclusion of untestable faults stated explicitly.

(h) **Fault simulation.** Predict faulty responses, measure coverage, identify undetected faults and grade/compact tests; optionally provide diagnostic signatures.



8 Switch-level XNOR and stuck-open tests

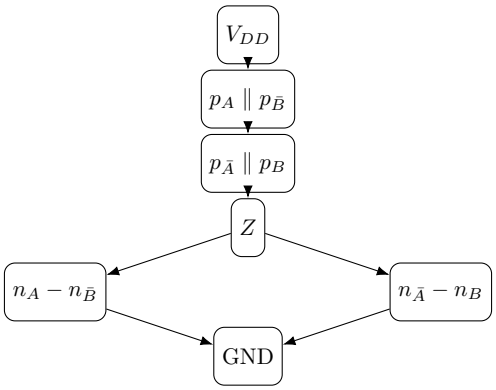
Question. Give a switch-level two-input XNOR. List all transistor stuck-open faults requiring two-pattern tests, with the tests.

Solution. Use two input inverters to supply $\bar{A}, \bar{B}$, and complementary pull-down/pull-up networks:

$$\text{PDN} = (n_A \text{ series } n_{\bar{B}}) \text{ parallel } (n_{\bar{A}} \text{ series } n_B),$$

$$\text{PUN} = (p_A \text{ parallel } p_{\bar{B}}) \text{ series } (p_{\bar{A}} \text{ parallel } p_B).$$

Subscripts denote gate signals. NMOS conducts for 1; PMOS for 0. This is an eight-transistor core plus four inverter transistors. The PDN realizes XOR, so the output realizes XNOR.



In the ideal charge-storage stuck-open model, initialize the output/internal node and then require the missing path:

Open transistor(s)	$(A, B)_{\text{init}} \rightarrow (A, B)_{\text{test}}$
p_A or p_B	$01 \rightarrow 00$
$p_{\bar{A}}$ or $p_{\bar{B}}$	$01 \rightarrow 11$
n_A or $n_{\bar{B}}$	$00 \rightarrow 10$
$n_{\bar{A}}$ or n_B	$00 \rightarrow 01$
PMOS in $\bar{A}$ inverter	$11 \rightarrow 01$
NMOS in $\bar{A}$ inverter	$01 \rightarrow 11$
PMOS in $\bar{B}$ inverter	$11 \rightarrow 10$
NMOS in $\bar{B}$ inverter	$10 \rightarrow 11$

For example $01 \rightarrow 00$ first discharges Z, then should charge it through p_A, p_B. If either is open, Z retains 0. For inverter opens, the first vector initializes the complementary signal. Charge sharing and leakage require electrical timing analysis beyond this model.